

Kaushik M

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EDUCATION

VIT Chennai

B.Tech in Computer Science Engineering

Chennai, India

2025 – 2029 | **CGPA: 9.13**

PROJECTS

Momenty | *Python*

- Built a monthly momentum strategy for the NIFTY 500 universe using K-Means clustering, Fama-French factor betas, and Efficient Frontier optimization
- Backtested strategy achieved **150% cumulative return vs. 65% benchmark** (2022–2026)

Variational Quantum Eigensolver | *Q#, Python*

- Implemented VQE with a custom two-layer ansatz built from scratch in Qiskit to estimate molecular ground-state energies
- Hydrogen benchmark: **-1.1077 Hartree vs. exact -1.1175 Hartree** (error ≈ 0.01 Hartree)

dctect | *Python*

- Built a detector for AI-generated images using Discrete Cosine Transform (DCT) frequency analysis with a full feature extraction and inference pipeline

ReelBloc | *Kotlin*

- Built an Android app using Accessibility Services to detect and block Reels/Shorts, helping users curb short-form video addiction

OptPricer | *C++*

- Built an option pricing engine implementing Black Scholes and Monte Carlo delta pricing methods

TECHNICAL SKILLS

Languages: Python, C, C++, Q#, SQL, Java, Kotlin, Rust

Quantum Computing: Qiskit, PySCF, OpenFermion, Azure QDK

Quant Finance & Algo Trading: PyPortfolioOpt, yFinance, statsmodels

ML / Data: TensorFlow, scikit-learn, NumPy, pandas, SciPy

Embedded Systems: ESP32, Arduino, Raspberry Pi

Infra: Kafka, Docker, Git, FastAPI, Flask

ACHIEVEMENTS

- LeetCode Knight – **Rating 1967** (Top 3% Globally)
- National Runner-Up – SP Robotics National Coding & Robotics Challenge (NCRC 2020)
- TIFA World Record – Microforest Plantation using Robotics, Anna University, 2020